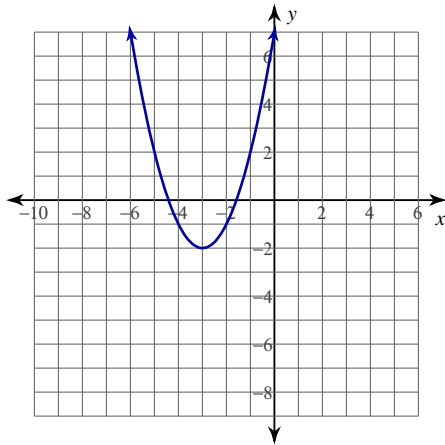


## Slope at a Value

Date \_\_\_\_\_ Period \_\_\_\_\_

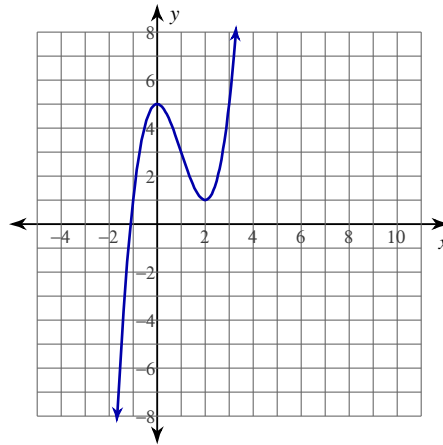
**For each problem, find the slope of the function at the given value.**

1)  $y = x^2 + 6x + 7$  at  $x = -2$



2

2)  $y = x^3 - 3x^2 + 5$  at  $x = 3$



9

3)  $y = x^3 - 6x^2 + 9x - 4$  at  $x = 2$

-3

4)  $y = -x^3 - 6x^2 - 9x + 1$  at  $x = -4$

-9

5)  $y = -\frac{1}{x^2 - 9}$  at  $x = 2$

 $\frac{4}{25}$ 

6)  $y = -\frac{3}{x + 5}$  at  $x = 1$

 $\frac{1}{12}$ 

7)  $y = -(x + 2)^{\frac{1}{3}}$  at  $x = -1$

 $-\frac{1}{3}$ 

8)  $y = -(-x + 2)^{\frac{1}{2}}$  at  $x = -5$

 $\frac{\sqrt{7}}{14}$ 

9)  $y = -\ln(-x + 2)$  at  $x = -3$

 $\frac{1}{5}$ 

10)  $y = \sin(2x)$  at  $x = -\pi$

2